When $a = 3$ and $b = 5$,

L.H.S = $(a+b)^2 = (3+5)^2 = 8^2 = 8\times 8 = \mathbf{64}$

R.H.S = $a^2 + 2ab + b^2 = 3^2 + (2\times 3\times 5) + 5^2 = 9 + 30 + 25 = \mathbf{64}$

Thus, for $a = 3$ and $b = 5$, L.H.S = R.H.S

Similarly, when $a = 4$ and $b = 7$,

L.H.S = $(a+b)^2 = (4+7)^2 = 11^2 = \mathbf{121}$

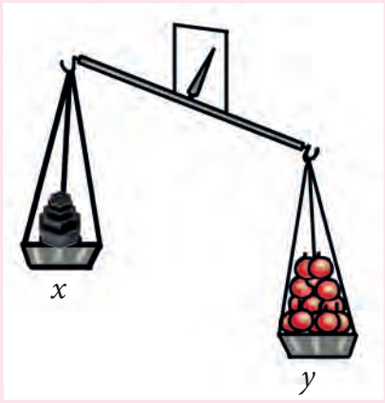
R.H.S = $a^2 + 2ab + b^2 = 4^2 + 2\times 4\times 7 + 7^2 = 16 + 56 + 49 = \mathbf{121}$

Also, for $a = 4$ and $b = 7$, L.H.S = R.H.S

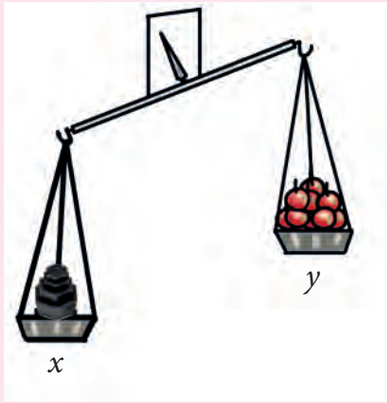
Thus, we shall find that for any value of 'a' and 'b' L.H.S = R.H.S. Such an equality, which is true for every value of the variable in it is called an identity. Thus, we observe that the equation $(a+b)^2 = a^2 + 2ab + b^2$ is an identity.

In general, algebraic equalities which hold true for all the values of the variables are called **Identities**. Let us see the basic identities with geometrical proof.

MATHEMATICS ALIVE - Inequalities in Real Life



$x < y$



$x > y$

3.2 Geometrical Approach to Multiplication of Monomials

We have already learnt that the operation of multiplication can be modelled in different ways. The one that we use in this unit is the representation of multiplication as a product table that is similar to area.

For example, the product 5×3 can be represented as shown in *Fig. 3.1*, which has five rows and three columns and it comprises 15 small squares. From *Fig. 3.2*, it is also clear that the product of 5×3 is the same as the product of 3×5 [Since, multiplication is commutative].

Here, multiplication is represented using grid model. The same can be represented using area model.

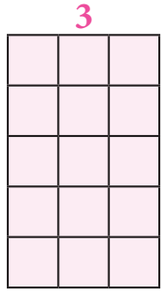


Fig. 3.1

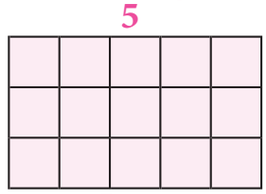


Fig. 3.2

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